Unraveling Ecological Drivers of Mammalian Macroevolution

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Introduction

- Body size is a central trait in ecology and evolution; clades tend to increase in size over macroevolutionary timescales (Alroy, 1998), but the drivers of this pattern, and how it varies with diet, remain poorly understood.
- The Eocene-Oligocene Transition (EOT, ~33 Ma) brought global cooling and aridification, driving the expansion of open, grass-dominated habitats at the expense of tropical/subtropical forests.
- Following the EOT, species show a gradual expansion of body sizes, leaving a gap in morphospace (~3–10 kg) separating small- and large-bodied lineages (Alroy, 1998).

How does the evolution of open landscapes influence the macroevolution of grazers, browsers, and mixed feeders?

Our goal is to connect landscape-scale changes to the ecological forces driving shifts in body size selection.

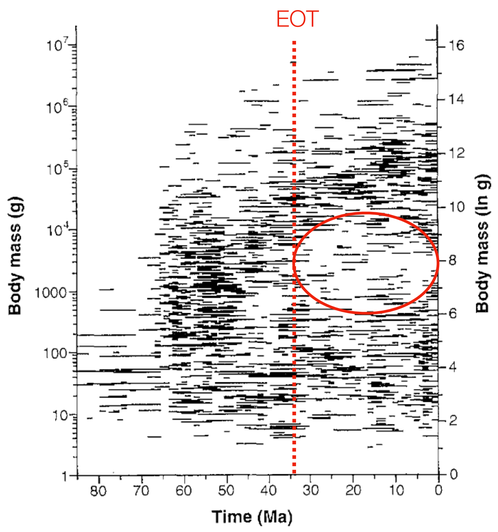
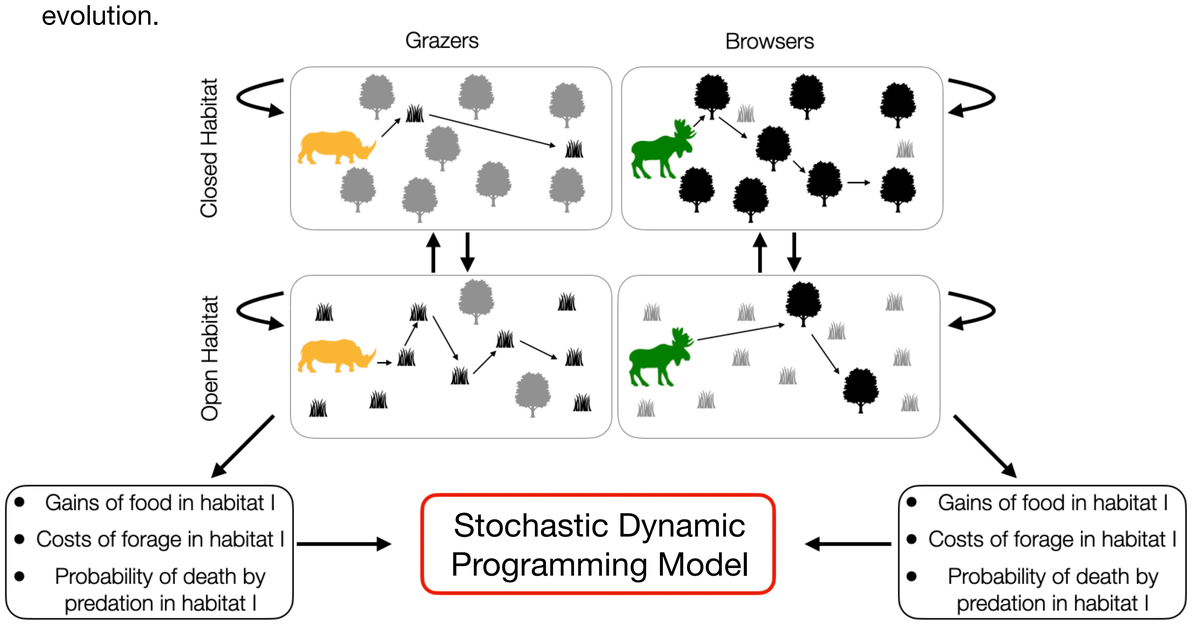


Figure adapted from Alroy (1998), Science 280(5364):731–734, with EOT boundary and the hole in morphospace highlighted

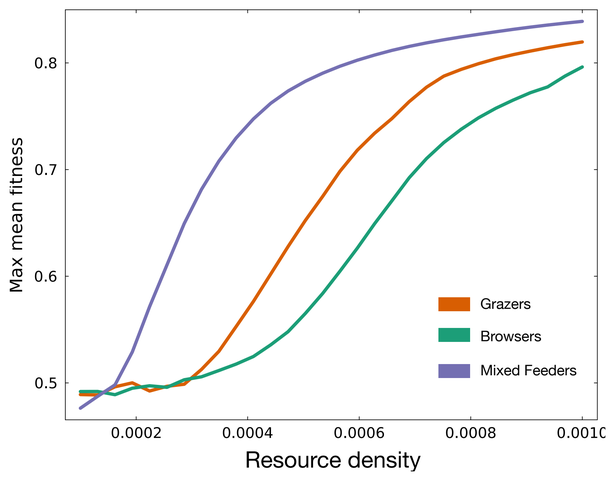
Methods

- We developed a bioenergetic model to calculate survival probability. The model includes an open habitat and a closed habitat. We use it to estimate an individual's survival over a period of time, given its body size and feeding strategy
- Each habitat is characterized by four key ecological parameters: the energetic cost of foraging, the energetic value of the food available, the spatial distribution of resources, and the probability of death by predation.
- These parameters differ between open and closed habitats, and between grazers and browsers — capturing the key ecological differences that we expect to matter for body size evolution.

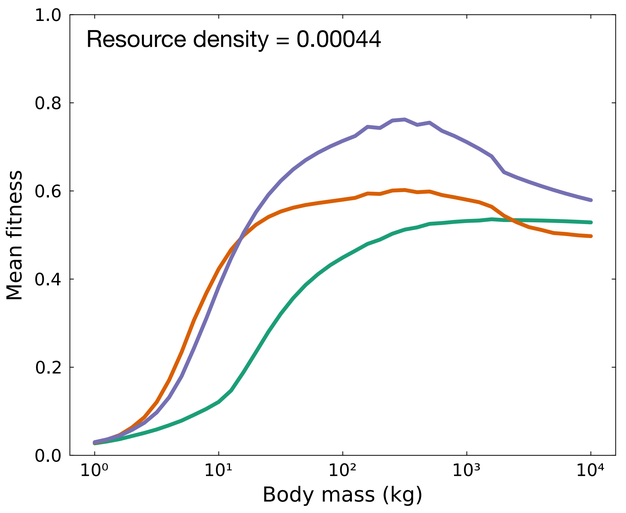


Results

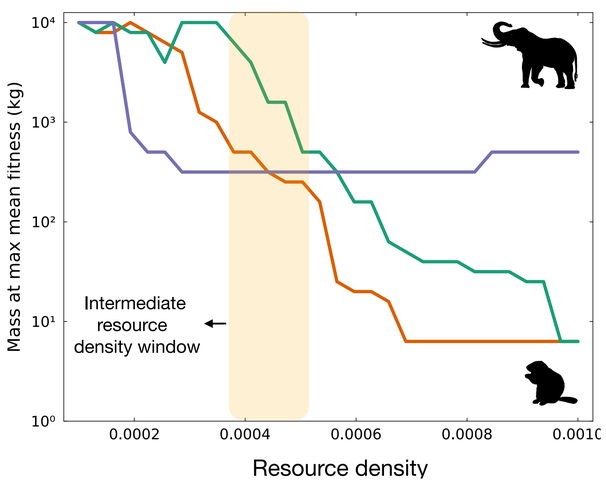
- Fitness increases with resource density, but the rate and shape of that increase differ between grazers, browsers, and mixed feeders. Mixed feeders show the highest fitness, followed by grazers and then browsers



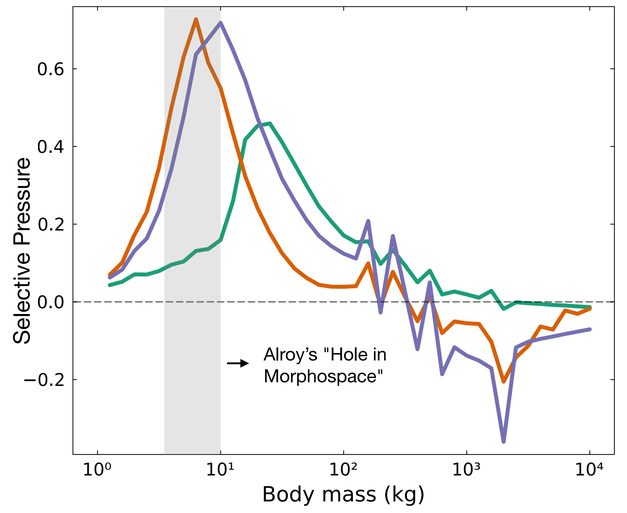
- At an intermediate value of resource density (0.00044), we observe that mixed feeders tend to have higher fitness than grazers, which in turn tend to have higher fitness than browsers.



- At low resource densities, larger animals have a fitness advantage, while smaller animals perform better in high resource density environments. Unlike browsers and grazers, mixed feeders appear less sensitive to this environmental pressure;
- At intermediate resource densities (yellow window), fitness-maximizing mass qualitatively aligns with their body mass modes post EOT



- Positive selective pressure is strongest in the 3–10 kg range, favoring larger sizes, consistent with the fossil evidence pattern of directional size increase among small-bodied herbivores after the EOT.



Conclusions

- Environmental conditions, such as resource density, together with adaptations for grazing or browsing, act as selective pressures shaping optimal body size in herbivores.
- Distribution of resources shapes both the fitness landscape and the intensity of selection across body sizes, with the strongest selective pressure concentrated at small body masses (~3–10 kg).
- Resource distribution fundamentally shapes both which body sizes are favored and how strongly selection acts on them. These results suggest that ecological context is a key driver of the macroevolutionary patterns we see in the fossil record.

References

Alroy, John. "Cope's rule and the dynamics of body mass evolution in North American fossil mammals." *Science* 280.5364 (1998): 731-734.Brown, J. H., Gillooly, J. F., Allen, A. P., Savage, V. M., and West, G. B. (2004). Toward a metabolic theory of ecology. Ecology, 85(7):1771–1789
Cerling, T. E., Harris, J. M., MacFadden, B. J., Leakey, M. G., Quade, J., Eisenmann, V., and Ehleringer, J. R. (1997). Global vegetation change through the miocene/pliocene boundary. Nature, 389(6647):153–158.402
Clauset, A. and Erwin, D. H. (2008). The evolution and distribution of species body size.Science, 321(5887):399–401
Clauss, M., Steuer, P., M¨uller, D. W., Codron, D., and Hummel, J. (2013). Herbivory and body size: allometries of diet quality and gastrointestinal physiology, and implications for herbivore ecology and dinosaur gigantism. PloS one, 8(10):e6871
Codron, D. (2019). Evolution of large mammal herbivores in savannas. Savanna woody plants and large herbivores, pages 213–243.
Damuth, J. (1981). Population density and body size in mammals. Nature, 290(5808):699-700.
Damuth, J. and Janis, C. M. (2011). On the relationship between hypsodonty and feeding ecology in ungulate mammals, and its utility in palaeoecology. Biological Reviews,86(3):733–758.417
Gordon, I. J. and Prins, H. H. (2008). The ecology of browsing and grazing. Number 195.Springer.
Mangel, M. and Clark, C. W. (1988). Dynamic modeling in behavioral ecology, volume 8.Princeton University Press.
Owen-Smith, R. N. (1988). Megaherbivores: the influence of very large body size on ecology.Cambridge university press.
Pound, M. J. and Salzmann, U. (2017). Heterogeneity in global vegetation and terrestrial climate change during the late eocene to early oligocene transition. Scientific Reports,7(1):43386.
Pringle, R. M., Abraham, J. O., Anderson, T. M., Coverdale, T. C., Davies, A. B., Dutton, C. L., Gaylard, A., Goheen, J. R., Holdo, R. M., Hutchinson, M. C., et al. (2023). Impacts of large herbivores on terrestrial ecosystems. Current Biology, 33(11):R584–R610.
Saarinen, J. (2019). The palaeontology of browsing and grazing. The ecology of browsing and grazing II, pages 5–59

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